WEEK 3 · LESSON 18 OF 24 · HANDOUT

Week 3 Project: Evaluate a Model

Study time: **about 60 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=LPZh9BOjkQs — open the lesson online to watch inside the course.

Your goal in this lesson: Evaluate a model's results like a professional: read the numbers, spot overfitting, pick the right measure for the decision, and write a clear recommendation.

About this project

You are a junior analyst at a health organisation in Rwanda. Two teams built models to predict which pregnant women are at **high risk** and should get extra clinic visits. Your manager asks: *"Which model should we use, and is it ready?"* Write a 1–2 page evaluation report.

The results

Both models were tested on the same **test set of 1,000 women**, of whom **100 are truly high-risk**.

Model A (a large neural network)

	Truly high-risk	Truly low-risk
Predicted high-risk	70	30
Predicted low-risk	30	870

Training accuracy: 99%. Validation accuracy: 91%.

Model B (a random forest)

	Truly high-risk	Truly low-risk
Predicted high-risk	88	112
Predicted low-risk	12	788

Training accuracy: 89%. Validation accuracy: 87%.

Step 1 — Calculate

For each model, calculate **accuracy, precision and recall**. Show your working.

Step 2 — Check for overfitting

Compare training and validation accuracy for each model. Which shows signs of overfitting? What would you suggest to the team?

Step 3 — Choose the measure that matters

- What does a **false negative** mean here? (A high-risk woman not given extra care.)

- What does a **false positive** mean? (A low-risk woman given extra visits.)
- Which mistake is more costly? So which measure matters most?

Step 4 — Recommend

Write a short recommendation (one paragraph): which model, why, and what must happen before it is used with real patients (for example, testing in more districts, checking fairness across rural and urban clinics, and a doctor always making the final decision).

Step 5 — One-slide summary

Make one slide or half-page summary a busy manager could read in one minute: the choice, three key numbers, and one risk.

Check your numbers

- Model A: accuracy 94%, precision 70%, recall 70%.
- Model B: accuracy 87.6%, precision 44%, recall 88%.

Model A has higher accuracy, but Model B catches far more high-risk women. In health screening, missing a high-risk woman is usually the more serious mistake — but Model B also sends many low-risk women for extra visits. Your report should weigh this trade-off honestly.

Remember this: Good evaluation is not "pick the highest accuracy". It means computing the right measures, checking for overfitting, linking errors to real costs and recommending clear next steps.

Name: _____ Date: _____

Activity

Task 1

Complete all five steps and write your report. Then add one paragraph: *"If clinics can only handle 150 extra-visit referrals per 1,000 women, how does this change your recommendation?"*

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Model A has TP = 70 and FN = 30. What is its recall?
   - A. 88%
   - B. 30%
   - C. 70%
   - D. 94%
2. Model B has TP = 88 and FP = 112. What is its precision?
   - A. 12%
   - B. 88%
   - C. 70%
   - D. 44%
3. Model A: training 99%, validation 91%. Model B: training 89%, validation 87%. Which shows more overfitting?
   - A. Model B
   - B. Both equally
   - C. Neither
   - D. Model A, because its gap is larger
4. In this health screening, what is a false negative?
   - A. A correct prediction
   - B. A high-risk woman not given extra care
   - C. A missing file
   - D. A low-risk woman sent for extra visits
5. Why is the highest accuracy not automatically the best choice?
   - A. Accuracy is always wrong
   - B. Accuracy cannot be compared
   - C. Accuracy ignores which kind of mistake is more costly
   - D. Managers dislike accuracy
6. What should happen before using the model with real patients?
   - A. Delete the test set
   - B. Nothing, deploy immediately
   - C. Remove all doctors
   - D. Wider testing, fairness checks and a doctor making the final decision

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